

M64 - Within-subject correlation: Q1-Q2 (order-aware) vs AH-PS (domain)

CORRELATIONS

Measure	r(Q1,Q2)	n	r(AH,PS)	n	diff
seq_typing_entropy	-0.075	71	-0.089	71	+0.015
seq_typing_mean_run_explore	-0.007	66	-0.000	66	-0.007
seq_typing_max_run	+0.042	71	+0.037	71	+0.006
first_writing_time_s	+0.261	71	+0.277	71	-0.016
final_answer_length	+0.694	71	+0.632	71	+0.061

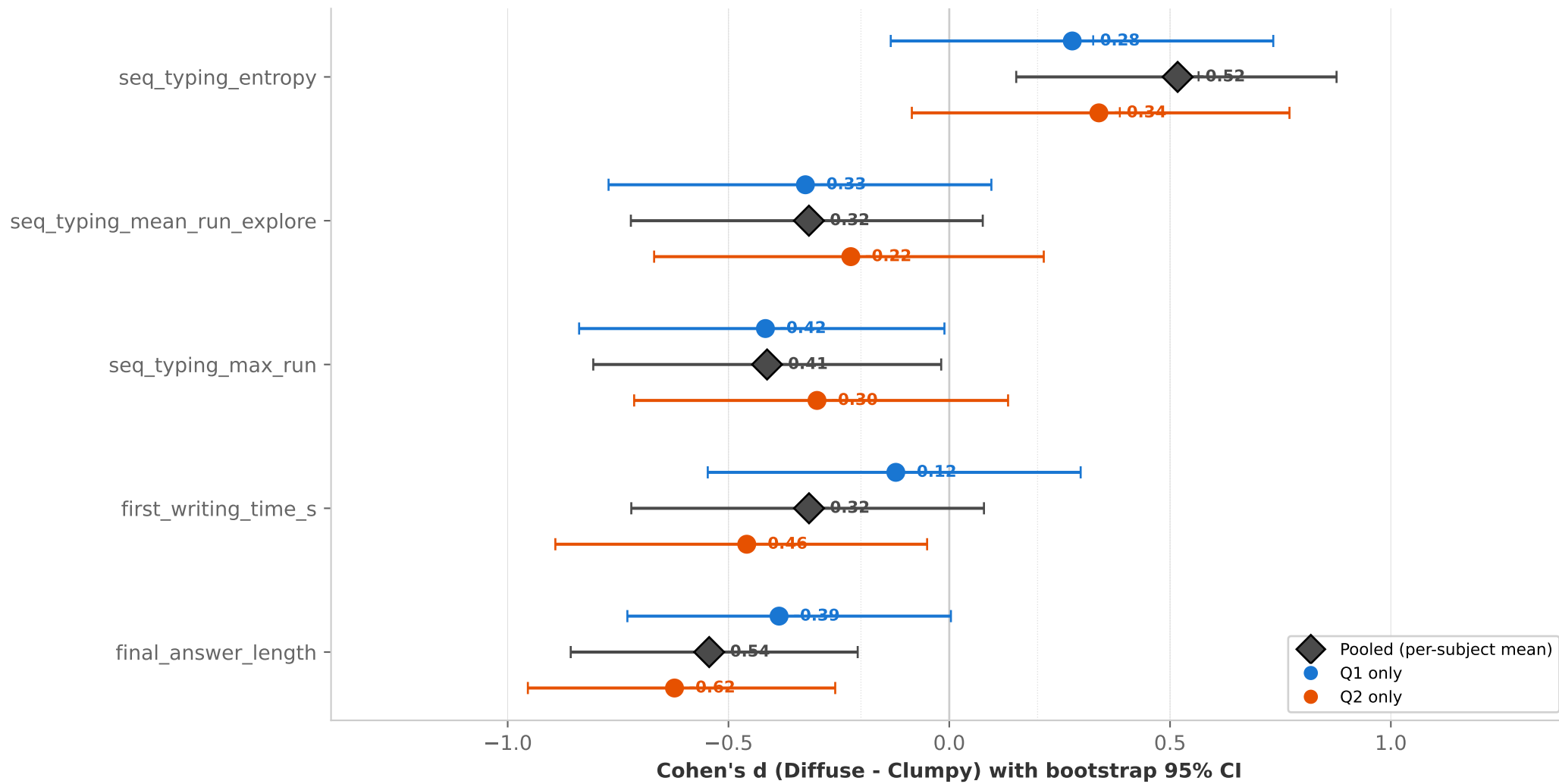
Yuval's hypothesis: if order effect drives the AH-PS zero correlation, then $r(Q1,Q2) - r(AH,PS)$ should be POSITIVE (Q1-Q2 more reliable).

ORDER EFFECT (paired t: Q2 - Q1)

Measure	mean(Q1)	mean(Q2)	diff	t	p	d_z
seq_typing_entropy	0.715	0.786	+0.071	-1.53	0.130	+0.18
seq_typing_mean_run_explore	3.688	3.975	+0.287	-0.52	0.602	+0.06
seq_typing_max_run	4.606	4.549	-0.056	+0.10	0.917	-0.01
first_writing_time_s	132.474	114.214	-18.260	+1.20	0.235	-0.14
final_answer_length	926.014	1156.127	+230.113	-2.62	0.011*	+0.31

A significant order effect supports Yuval's hypothesis (warming up, practice, fatigue) - it suggests stratifying or controlling for order.

M64 - Bootstrap effect sizes by trial position



M64 - Bootstrap Cohen's d (5000 resamples)

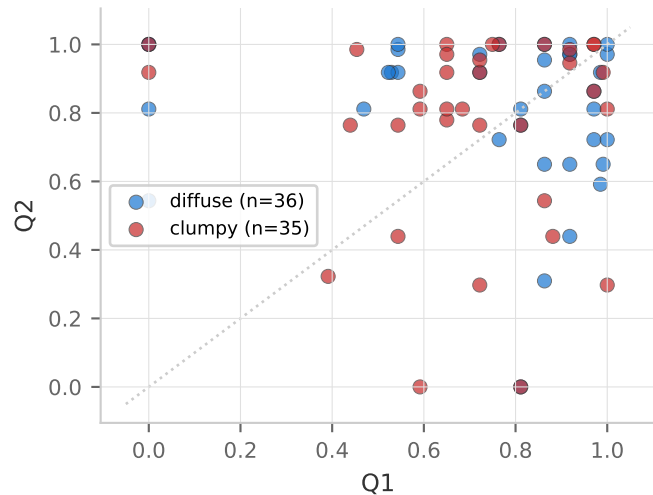
Measure	Stratum	N(D/C)	d	95% CI	p
seq_typing_entropy	pooled	52/49	+0.52	[+0.15, +0.88]	0.009
	pos1	46/44	+0.28	[-0.13, +0.73]	0.196
	pos2	42/40	+0.34	[-0.08, +0.77]	0.118
seq_typing_mean_run_explore	pooled	52/47	-0.32	[-0.72, +0.08]	0.113
	pos1	43/42	-0.33	[-0.77, +0.10]	0.135
	pos2	42/38	-0.22	[-0.67, +0.21]	0.310
seq_typing_max_run	pooled	52/49	-0.41	[-0.81, -0.02]	0.040
	pos1	46/44	-0.42	[-0.84, -0.01]	0.045
	pos2	42/40	-0.30	[-0.71, +0.13]	0.158
first_writing_time_s	pooled	52/49	-0.32	[-0.72, +0.08]	0.111
	pos1	46/44	-0.12	[-0.55, +0.30]	0.569
	pos2	42/40	-0.46	[-0.89, -0.05]	0.028
final_answer_length	pooled	52/49	-0.54	[-0.86, -0.21]	0.003
	pos1	46/44	-0.39	[-0.73, +0.00]	0.053
	pos2	42/40	-0.62	[-0.95, -0.26]	0.002

p = bootstrap directional p (2 * min of tails crossing zero).

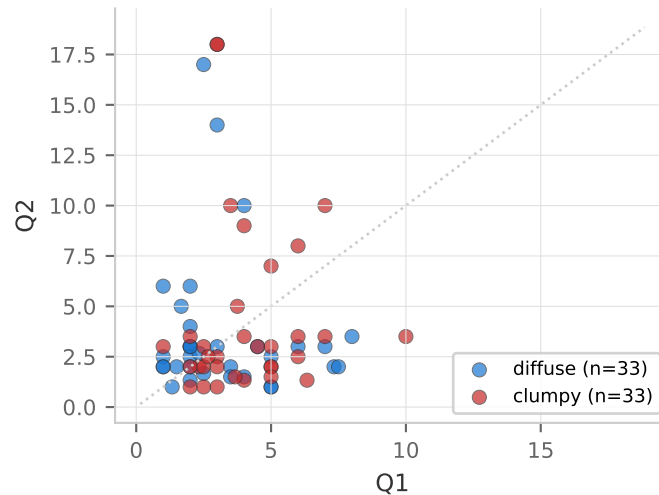
CI excluding 0 = effect coherent across the group.

M64 - Within-subject Q1 vs Q2 scatters

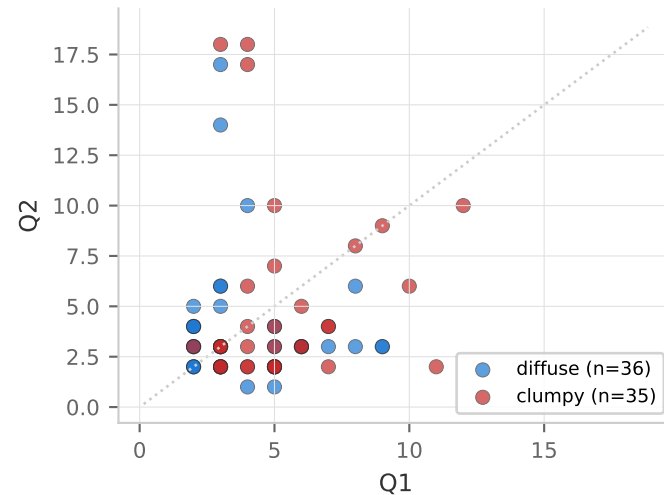
seq_typing_entropy
 $r(Q1, Q2) = -0.07$ $p = 0.536$



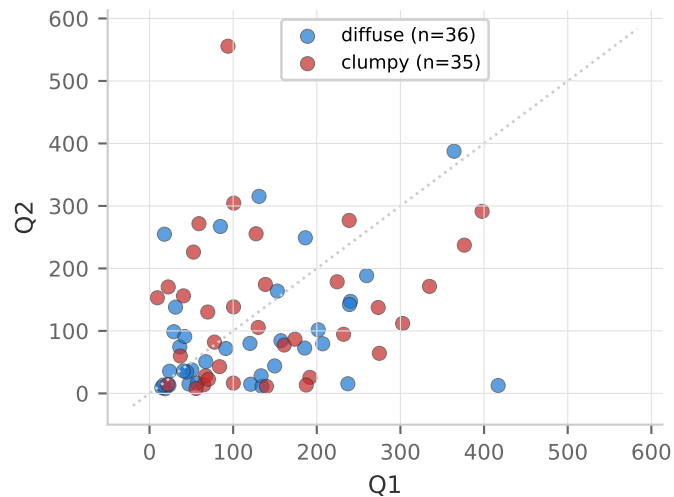
seq_typing_mean_run_explore
 $r(Q1, Q2) = -0.01$ $p = 0.956$



seq_typing_max_run
 $r(Q1, Q2) = +0.04$ $p = 0.727$



first_writing_time_s
 $r(Q1, Q2) = +0.26$ $p = 0.028$



final_answer_length
 $r(Q1, Q2) = +0.69$ $p = 0.000$

